MAHATMA GANDHI UNIVERSITY, KOTTAYAM
Semester IV - Industrial Chemistry Minor Practical Examination
Model Question Paper
Course Code: MG4DSCICH200
Course Title: Instrumentation Techniques in Chemistry

Time: 3 Hours **Maximum Marks: 35**

Part – I (Answer only the specified question, 5 marks)

1. Identify and interpret the functional groups present in the given IR spectrum of an organic compound.
2. Measure the refractive index of the given liquid using an Abbe refractometer and report the value.
3. Perform calibration of any one analytical instrument, suggested by the examiner.
(UV–Visible Spectrophotometer / Refractometer / Polarimeter / Conductivity Meter / Potentiometer)

Part – II (Answer only the specified question, 5 marks)

3. Write the brief procedure for the experiment in question number ..... from the Part III in the first 10 minutes.

Part – III (Answer only the specified question, 10 marks)

4. Determine the unknown concentration of the given sample of Ferric ion using UV–Visible Spectroscopy / Colorimeter .
5. Verify Beer–Lambert's Law using UV–Visible Spectroscopy for the given solution.
6. Determine the concentration of the given sugar sample using refractometry.
7. Measure the optical rotation of the given solution using a polarimeter and calculate the specific rotation.
8. Calculate the strength of given strong acid by performing conductometric titration against a strong base.
9. Find out the cell constant of the given conductivity cell and hence calculate the conductivity of the given electrolyte.
10. Perform potentiometric titration of ferrous ion against standard potassium permanganate solution and hence find out the concentration of ferrous ion.
11. Determine the strength of given weak acid by performing conductometric titration against strong base.

Part – IV

12. Viva Voce – **10 marks**
13. Lab Record – **5 marks**